

7.12.2019 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov- Dec 2019

Duration: 2 Hrs
Semester: V

Max. Marks: 50

Class: TYBTech

Name of the Course: Software Engineering

Branch: Computer

Course Code: UCEC505 PWD Students Section A

Instructions:

- (1) All Questions are Compulsory
- (2) Assume suitable data if necessary

Question No.	Section A	Max. Marks
Q 1 (a)	Differentiate between user and system requirements? State any three differences – 2 M each	6
Q 1 (b)	Define Coupling and Cohesion Coupling Definition – 2 M Cohesion Definition – 2 M	4
Q 1 (c)	Explain different types of coupling and Cohesion. Three types each – 1 mark each	6
Q2 (a)	If you have to develop a word processing software product, what process model will you choose? Justify. Explain the process model Stating correct process model – 2 M Justification – 2 M Description of process model – 4 M OR <i>explain any four non-functional requirements</i>	8
Q2 (b)	Explain the steps involved in project planning OR Discuss about COCOMO II model for software estimation	8
Q 2(c)	Draw class diagram for “Attendance monitoring system using OTP” State different classes and their relationships. Also define multiplicity along relation OR Draw use case diagram for any one scenario of “Automated Teller Machine” State different users participating in that scenario. Show include relation in the use case	8

Q3 (a)

1. Which one of the following models is not suitable for accommodating any change?
 - a) Build & Fix Model
 - b) Prototyping Model
 - c) RAD Model
 - d) Waterfall Model
2. Which one of the following is not a phase of Prototyping Model?
 - a) Quick Design
 - b) Coding
 - c) Prototype Refinement
 - d) Engineer Product
3. Which model can be selected if user is involved in all the phases of SDLC?
 - a) Waterfall Model
 - b) Prototyping Model
 - c) RAD Model
 - d) both Prototyping Model & RAD Model
4. Which one of the following is a functional requirement ?
 - a) Maintainability
 - b) Portability
 - c) Robustness
 - d) None of the mentioned
5. "Consider a system where, a heat sensor detects an intrusion and alerts the security company." What kind of a requirement the system is providing ?
 - a) Functional
 - b) Non-Functional
 - c) Known Requirement
 - d) None of the mentioned
6. Which one of the following is not a step of requirement engineering?
 - a) elicitation
 - b) design
 - c) analysis
 - d) documentation
7. Classes and interfaces are a part of
 - a) Structural things
 - b) Behavioral things
 - c) Grouping things
 - d) Annotational things

10 (1 M each)

8. Which diagram shows the configuration of run-time processing elements?

- a) Deployment diagram
- b) Component diagram
- c) Node diagram
- d) ER-diagram

9. _____ allows us to infer that different members of classes have some common characteristics.

- a) Realization
- b) Aggregation
- c) Generalization
- d) dependency

10. _____ & _____ diagrams of UML represent Interaction modeling.

- a) Use Case, Sequence
- b) Class, Object
- c) Activity, State Chart
- d) All of the mentioned

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Max. Marks: 50**Class: TYBTech****Name of the Course: Software Engineering****Branch: Computer****Course Code: UCEC505 PWD Students Section B****Duration: 2 Hrs****Semester: V****Instructions:**

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Question no	Section B	Max. Marks
Q4 (a)	A program specifications state the following for the input field: The program shall accept an input value of 4 digit integer equal or greater than 2000 and less than or equal 8000. Determine the test cases using: • Equivalence class partitioning - 4 M • Boundary value analysis - 4 M OR Explain the process of Unit testing and Integration Testing 4 M each	8
Q 4 (b)	Explain User interface Design process? 5 M Draw graphical user interface for "Restaurant table booking" system 5 M	10
Q 4 (c)	Explain the elements used in sequence diagram. 4 M Draw sequence diagram for the software system-"Ordering Pizza over phone". Consider the scenario from the moment you pick up the phone to the point you start eating pizza 6 M OR Explain the elements used in Activity diagram. 4 M Construct activity Diagram for the software system-"Ordering Pizza over phone". Consider the scenario from the moment you pick up the phone to the point you start eating pizza 6 M	10
Q 5 (a)	Explain Change control and version control management – 3 M each	6

Q 5 (b)	List different types of architectural styles – 2 M Explain any one architectural style – 4 M OR Describe the golden rules for interface design?	6
Q6	1. Which of the following is golden rule for interface design? a) Place the user in control b) Reduce the user's memory load c) Make the interface consistent d) All of the mentioned 2. What do you understand by V&V in software testing? a) Verified Version b) Version Validation c) Verification and Validation d) Version Verification 3. Which granularity level of testing checks the behavior of module cooperation? a) Unit Testing b) Integration Testing c) Acceptance Testing d) Regression Testing 4. Which of the following is not a direct measure of SE process? a) Efficiency b) Cost c) Effort Applied d) All of the mentioned 5. Software mistakes during coding are known as a) errors b) failures c) bugs d) defects 6. The modification of the software to match changes in the ever changing environment, falls under which category of software maintenance? a) Corrective b) Adaptive c) Perfective d) Preventive 7. Which of the following is/are commonly used architectural pattern(s)? a) Model-View-Controller b) Layered Architecture c) Client-server d) All of the mentioned	10

8. In which of the following testing strategies, a smallest testable unit is the encapsulated class or object?
- a) Unit testing
 - b) Integration testing
 - c) System testing
 - d) None of the mentioned

9. Coding is not design, design is not coding:
- a) True
 - b) False

10. Systems that involve interaction with a shared database can be considered as.
- a) software-based
 - b) transaction-based
 - c) server-based
 - d) client-based